🌈 Colour Pillars — English Worksheet

Topic: Six variables and `+ 4` · **Course:** 3D Coordinates · **Level:** Extension · **Time:** about 40 minutes

WORLD: VARIABLES INTRO · TYPE `DEL` BEFORE EVERY `RUN`. EVERY TIME.

Every `???` is one short thing for you to work out. There are no answers on this sheet.

Count on the screen. Do not count in your head.

1 · Six variables. That is all 🎯

You get these six. No new ones.

```
x_start = 2
y_start = 0
z_start = 2
x_end = 2
y_end = 3
z_end = 2
```

One `fill` from `y_start` to `y_end`. How many blocks tall is that? Count 0, 1, 2, 3.

Why is `y_end` 3 and not 4?

A pillar is three of these stacked up. Each part is a different block.

Last week you made a new variable for every part. Why is that slow?

2 · Plus 4, minus 4 ➕➖

You do not need new variables. You do maths inside `pos()`.

```
pos(x_start, y_start + 4, z_start)
```

Fill the table. Part 1 is done for you.

PART	BLOCK	FIRST Y	LAST Y	MATHS YOU WRITE
1	bottom	0	3	y_start y_end
2	middle			
3	top			

Why do you add 4 and not 3?

Now start from the top instead.

If the top part uses `y_start`, what maths gives you the middle part and the bottom part?

3 · One pillar, three colours 🌈

Pick three blocks you like. Write the block name straight inside `fill`.

TEMPLATE

```
blocks.fill(REDSTONE_BLOCK,
    pos(x_start, y_start, z_start),
    pos(x_end, y_end, z_end), REPLACE)
blocks.fill(GOLD_BLOCK,
    pos(x_start, y_start + ???, z_start),
    pos(x_end, y_end + ???, z_end), REPLACE)
blocks.fill(DIAMOND_BLOCK,
    pos(x_start, y_start + ???, z_start),
    pos(x_end, y_end + ???, z_end), REPLACE)
```

CAREFUL

Both corners of one part move by the same amount. Move only the end and the part gets longer instead of moving up.

Run it. How tall is your whole pillar? Count it on the screen.

WORKS WHEN

you see three colours, four blocks each, standing in one straight pillar.

TWO COLOURS TOUCHING IN A STRANGE PLACE?

Your maths is `+ 3` or `+ 5`. Count one part again: 0, 1, 2, 3.

4 · A row of pillars 🔁

Now build four pillars in a row. Only `z` changes.

TEMPLATE

```
def on_chat():
    x_start = 2
    y_start = 0
    z_start = 2
    x_end = 2
    y_end = 3
    z_end = 2
    for i in range(???):
        blocks.fill(REDSTONE_BLOCK,
            pos(x_start, y_start, z_start),
            pos(x_end, y_end, z_end), REPLACE)
        blocks.fill(GOLD_BLOCK,
            pos(x_start, y_start + ???, z_start),
            pos(x_end, y_end + ???, z_end), REPLACE)
        blocks.fill(DIAMOND_BLOCK,
            pos(x_start, y_start + ???, z_start),
            pos(x_end, y_end + ???, z_end), REPLACE)
        z_start = z_start + ???
        z_end = z_end + ???
player.on_chat("pillars", on_chat)
```

CAREFUL

The two `z` lines go **inside** the loop, after all three fills. Count the spaces at the front.

Before you run: how many pillars, and where does the last one stand?

ALL FOUR PILLARS IN ONE SPOT?

Your `z` lines are outside the loop, so they run once at the end.

PILLARS STUCK TOGETHER?

`z` is going up by less than 4. Leave a gap and look again.

Your last pillar ends at z = ??? and y = ???. Write the `del` box that clears them all.

```
blocks.fill(AIR,
    pos(0, 0, 0), pos(???, ???, ???), REPLACE)
```

5 · Show Your Work 📷 🎥

Record **one video**, one take, no stopping. A phone is fine. Show these in order:

> 1. Type `del`, then `pillars`. Walk past all four pillars.
> 2. Point at your six variables and read them out.
> 3. Show one `+ 4` and say which part of the pillar it moves.
> 4. Show that the `z` lines are inside the loop.
> 5. Read row 3 of your table out loud.

Try to use these words: **variable**, **loop**, **start**, **end**, **plus four**.

Plan what you will say before you record.

Submit ✅

Send this worksheet and your video to teacher on KakaoTalk.